Stock Focus

The Lottery Corporation (TLC-ASX)

E&P

Powerball or bust

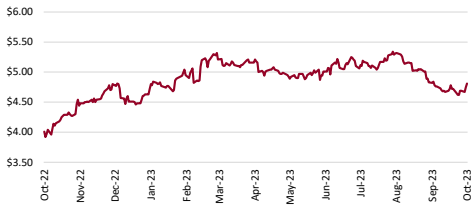
Recommendation: Neutral

Last price	**AUD4.81**
Valuation	**AUD5.40**

TSR

Upside to valuation	12.3%
Dividend yield	3.3%
Expected total SH return	15.6%

TLC-ASX share price history
Source: E&P Research, IRESS



Trading Data

Last price	AUD4.81
Valuation	AUD5.40
12 month range	AUD3.92 - AUD5.32
Market cap ($m)	$10,706
Shares outstanding (m)	2,225.8
Free float (%)	100.0
12 month return (historical)	14.4%

Earnings revisions

	FY24	% chg	FY25	% chg
Revenue	3,785.4	211.0%	3,945.2	171.6%
EBITDA	779.8	125.1%	819.3	86.3%
Adj. EPS	16.9	(78.2)%	18.0	(82.4)%
DPS	0.2	(70.6)%	0.2	(77.7)%
Val	**AUD5.40**	**-3.8%**		

Paul Mason
paul.mason@eandp.com.au
+61 3 9631 9867

Sam Bradshaw
sbradshaw@eandp.com.au
+61 3 9235 9706

Powerball carrying the day while other games lag

This note could be considered part-two of our initiation note Burger with the Lott. Many of the foundation concepts and mathematics explained in that note will be necessary to understand the topics we explore. The major elements of the report involve an analysis of:

- Powerball - looks like it is performing well, and is currently experiencing quite normal luck with solid like-for-like entries on higher ticket pricing;
- Oz Lotto - is off to a weak start to the year with bad luck continuing and a heavily depleted prize reserve fund that will require sustained slow sequences to rebuild;
- Saturday Lotto - record FY23 turnover was driven by some prize reserve fund depletion and also jackpot timing and presents a tough comp. Otherwise, like-for-like entries are not driving significant growth either.
- Overall portfolio is becoming more and more dependent on Powerball given entries mix shift over time.

Changes to forecasts and valuation

We have reduced our FY24-26 forecasts for EBITDA by -2.4%/-3.4%/-3.4% from further opex inclusions alongside minor adjustments to revenue forecasts. Our forecasts for 1H24 are 4-5% below consensus presently.

Our view

TLC is a generally high quality business, albeit with a luck factor that introduces volatility. Volatility continues to increase on a theoretical basis over time as turnover skews more towards Powerball. We see good valuation upside at current levels, albeit we believe consensus estimates are too high heading into the 1H24 result at the moment unless TLC goes through a high-luck streak for the rest of the half year. We remain Neutral given consensus positioning, but as will likely often be the case, still encourage long-term oriented investors to look at the stock generally favourably.

Earnings Forecasts

Yr to Jun (AUD)	23A	24E	25E	26E
EBITDA	713.2	779.8	819.3	856.2
Adjusted Net Profit	264.8	377.0	400.0	425.4
Adj. EPS (c)	11.9	16.9	18.0	19.1
PER (x)	**40.4**	**28.4**	**26.8**	**25.2**
Yield (%)	**3.1**	**3.3**	**3.3**	**3.3**
EV/EBITDA (x)	**17.9**	**16.1**	**15.2**	**14.4**
Valuation (blended)				**AUD5.40**

Source: E&P Research

Powerball or bust

In this report we provide some deeper analysis of what is currently going on across TLC's three largest games, Powerball, OzLotto and Saturday Lotto.

At a high level the report runs through the following:

- Powerball: has been performing well this year, and is experiencing slightly above average luck (YTD outcomes + a 50th percentile remainder of the year will lead to a 52st percentile outcome for FY24 on our estimates). We believe there is some evidence that the entries premium for $100m+ draws might be diminishing over time (ie. the shock value is gradually fading). However, the recent game change has been a relative success for the company with entries basically retained despite price rises offering the potential to accelerate the sequence on a more frequent basis;
- OzLotto: has been performing poorly this year, due to a combination of continued unfavorable luck (we believe YTD + a 50th percentile remainder of the year will lead to a 34th percentile outcome, i.e. really bad luck again to start FY24), but also headwinds on a like-for-like entries basis. On our calculations, TLC has heavily depleted the prize reserve fund for OzLotto this year and is needing to maintain a slow sequence for some time to rebuild;
- Saturday Lotto: is comping a tough FY23 where TLC used the prize reserve fund in assisting revenue generation, and also featured two $30m mid-year Megadraws that both fell in the same financial year. Furthermore, we are not seeing any signs of growth in like-for-like draws.

Overall we also suspect the success of Powerball as a brand is overwhelming the other TLC games, as its growth rates the last three years have substantially oustripped the other games. This hit a real turning point coming out of the lockdowns in particular, where as a portfolio the 'ex-Powerball' group of games have basically not grown, while Powerball has been on a tear. This is leading to a circumstance (at least potentially) where TLC's earnings will end up being more volatile and dependent on luck outcomes related to Powerball through time.

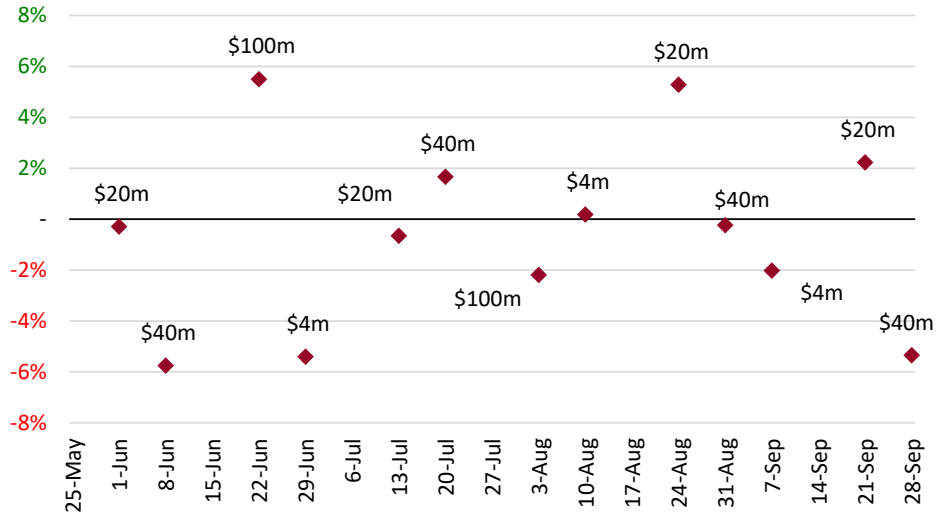
We like TLC as a business in terms of its general monopolistic characteristics and longer term growth, however remain at Neutral for now. We are sitting 4-5% below consensus for 1H24 results (we apply what we believe are statistical 50th percentile luck estimates with a bit of growth to all forecasts which means less variation longer term), and we believe consensus has not understood the depths of bad luck and entries headwinds for the ex-Powerball portfolio this year, particularly for OzLotto. Consequently, we continue to wait for a moment where we don't see downside earnings risk to consensus before becoming more Positive. For longer term investors that can look through shorter term earnings we would note that there is decent upside implied in our present $5.40/sh valuation.

Powerball - Showing strength post game change

Like-for-Like Entires

Below we have tracked the like-for-like jackpot entries since the May-23 game change against the 12 months pre game change. Despite the increase in subscription price and retailer commission (+11.34% overall), like-for-like entries remain robust. We had previously forecast a minor decline in like-for-like entries due to a combination of reasons: (i) higher entry price for the same division-one prize, (ii) increased performance from Oz Lotto causing some cannibalization; and (iii) macroeconomic impacts such as the rising cost of living.

Exhibit 1 - Change in Powerball Like-for-Like Jackpot Entries post May-23 game change vs. the 12 months pre game-change



Source: Company Data, E&P Estimates. Note: excludes jackpots with fewer than two data points in the prior period. Comments are like-for-like jackpot size

August-2023 $100m Draw

The recent $100m draw in August was the eleventh in history and fifth within the past 12 months. Estimating $100m+ draws accurately is important given the significant impact they have on the prize reserve fund and therefore the speed of the sequence that Powerball is capable of sustaining. We had previously endeavored to identify the factors that could help predict the number of entries that $100m + jackpots would generate. Our method used linear regression from smaller draws, extrapolated to the $100m+ jackpot size and then applied a premium. For the most recent draw, the premium was +20.1% which was the second lowest of the historic sample size.

Exhibit 2 - Key metrics that impact the Premiums from $100m+ Powerball Jackpots

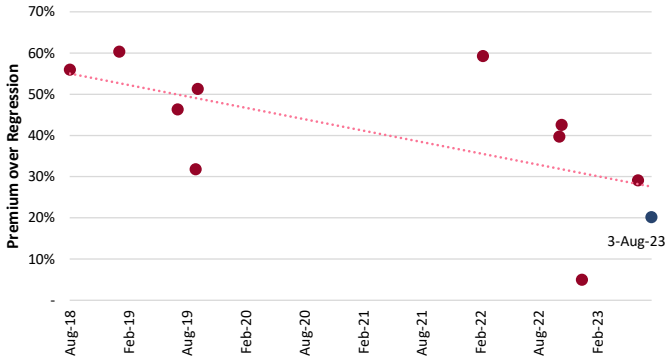
Timeframe		Result				Factors				
FY	Date	Jackpot	Actual Entries	Linear Estimation (a)	Premium	Days since 100m+ draw (b)	Largest in History (at time)	Sales Price	Next SL Jackpot	Prior OZL Jackpot
FY19	16-Aug-18	100,000,000	169,384,236	108,613,300	56.0%	na (c)	Yes	$1.20	4,359,646	2,000,000
FY19	17-Jan-19	100,000,000	174,145,592	108,613,300	60.3%	154	No	$1.20	4,118,836	10,000,000
FY20	18-Jul-19	110,000,000	149,659,729	102,276,000	46.3%	182	No	$1.20	4,316,189	10,000,000
FY20	12-Sep-19	100,000,000	124,361,859	94,361,000	31.8%	56	No	$1.20	3,841,185	5,000,000
FY20	19-Sep-19	150,000,000	202,664,967	133,936,000	51.3%	63	Yes	$1.20	4,164,796	10,000,000
FY22	24-Feb-22	120,000,000	244,631,645	153,613,000	59.3%	889	No	$1.21	10,000,000	2,000,000
FY23	20-Oct-22	100,000,000	183,084,473	131,034,000	39.7%	238	No	$1.21	5,024,463	5,000,000
FY23	27-Oct-22	160,000,000	284,837,414	199,770,000	42.6%	245	Yes	$1.21	20,000,000	10,000,000
FY23	29-Dec-22	100,000,000	137,554,844	131,034,000	5.0%	63	No	$1.21	40,000,000	10,000,000
FY23	22-Jun-23	100,000,000	169,147,583	131,034,000	29.1%	175	No	$1.35	30,000,000	3,000,000
FY24	3-Aug-23	100,000,000	156,823,488	130,525,000	20.1%	42	No	$1.35	20,000,000	10,000,000

a) linear regression is ex >=100m draws
b) excludes >=$100m jackpots in the same sequence
c) first 100m jackpot for PBL

Source: Company Data, E&P Estimates. Note: Includes WA

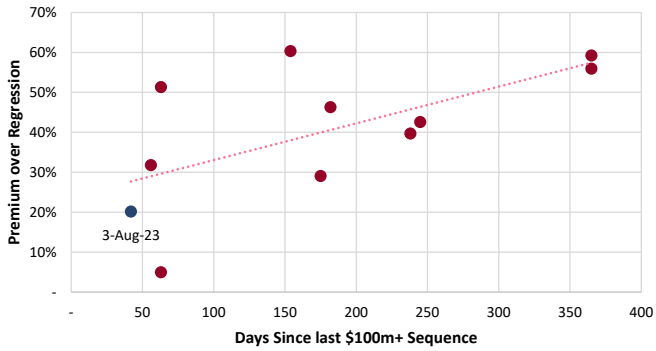
We suspect that the mystique of the $100m+ draw has been declining over time. We also believed that entries would be weaker if another $100m+ draw had occurred fairly recently (i.e. jackpot fatigue). In fact, the Aug-23 jackpot was the second of back-to-back sequences that reached $100m, resulting in -7.3% less entries than the $100m draw six weeks prior. The other factor to consider was that the Saturday Lotto draw two days later was a $20m Superdraw, likely leading to more cannibalistion than if it were normal draw.

Exhibit 3 - $100m+ Powerball Jackpot Premium over Time



Source: Company Data, E&P Estimates. Note: Includes WA.

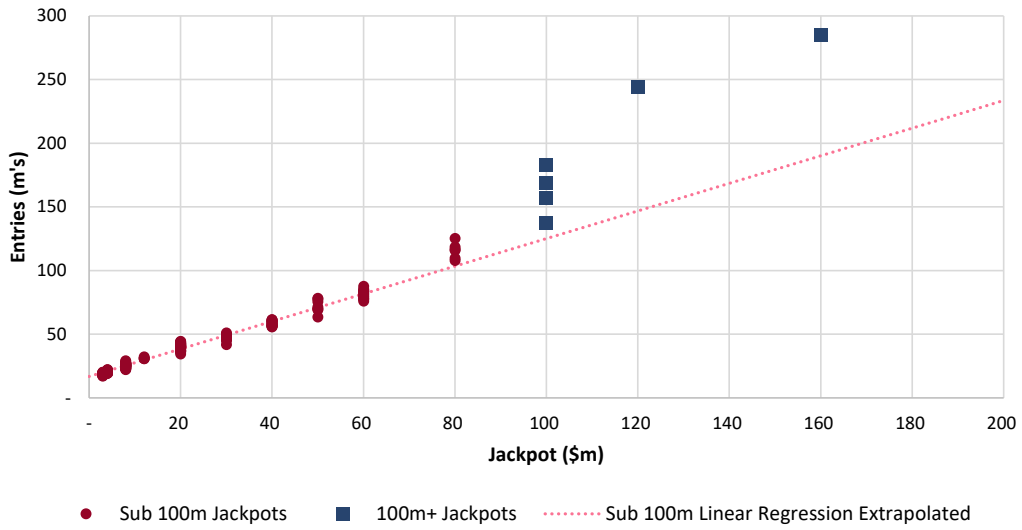
Exhibit 4 - $100m+ Powerball Jackpot Premium based on the number of days since most recent $100+m Jackpot



Source: Company Data, E&P Estimates. Note: Includes WA. Note: Last $100m+ jackpot does not include jackpots in the same sequence. Days are capped at 365.

Our prior work has noted the shape of the trend in entries vs jackpot size, with what looks like an S-curve that accelerates around the $100m mark, and a generally linear trajectory for lower jackpots. The premium is therefore defined by how big a jump in entries occurs vs. the otherwise trend line for $100m+ draws.The chart below illustrates this historical premium that has occurred vs. the linear trend once jackpots get to $100m. In the prior charts we have shown how this premium appears to be declining through time, albeit on a relatively low sample size.

Exhibit 5 - Powerball Entries from sub $100m Jackpots - Linear Regression Exprapolation

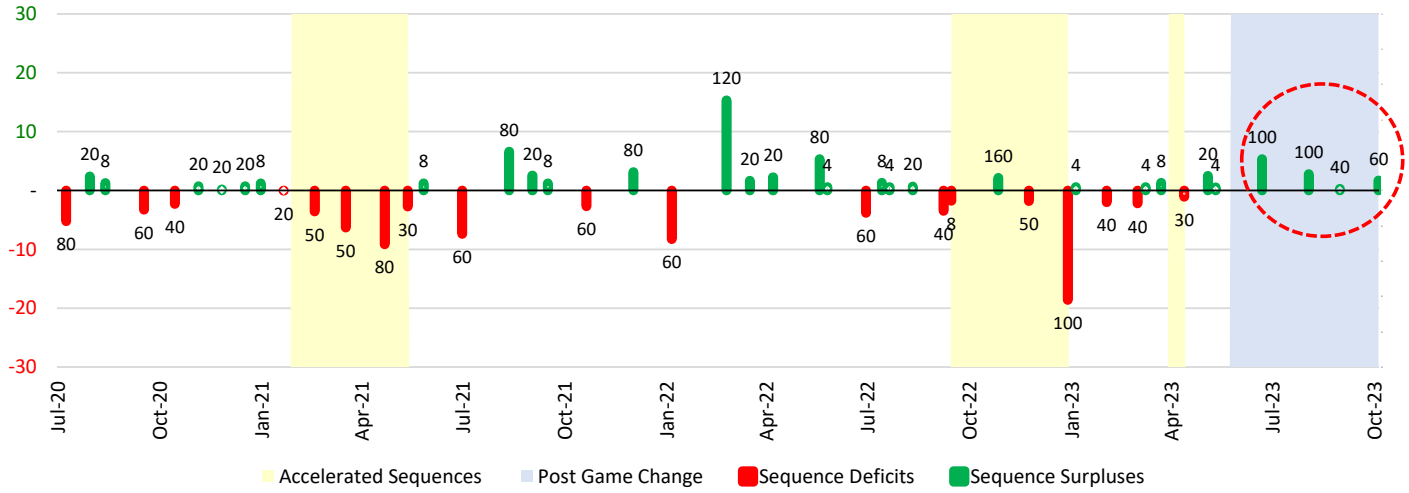


Source: Company Data, E&P Estimates. Note: Includes WA. Data is from FY21

Prize Reserve Fund post Game Change

Despite the declining premiums, the $100m+ draws are still very positive for the prize reserve fund in the current format post game change. Furthermore, the most recent $40m & $60m sequences also had a positive sequence surpluses despite the $40m recently having sequence deficits for the same sequence.

Exhibit 6 - Powerball Net Sequence Prize Reserve Fund Accumulation (A$m) - jackpot as label (A$m)

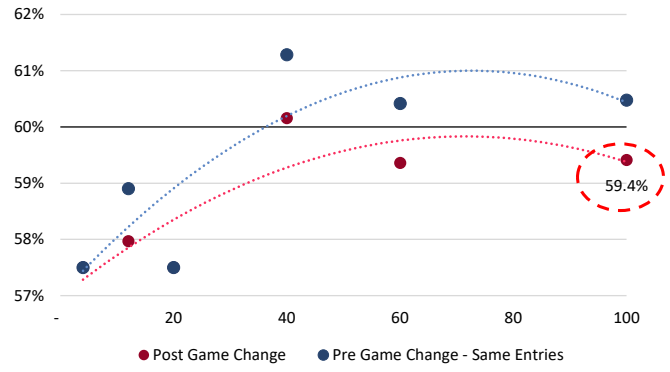


Source: Company Data, E&P Estimates. Note: Includes WA

The reason for this is the impacts from the May-23 game change, particularly the increase in the subscription price which we previously explored in Burger with the Lott. We demonstrate this below by comparing the actual Aug-23 $100m jackpot sequence against a

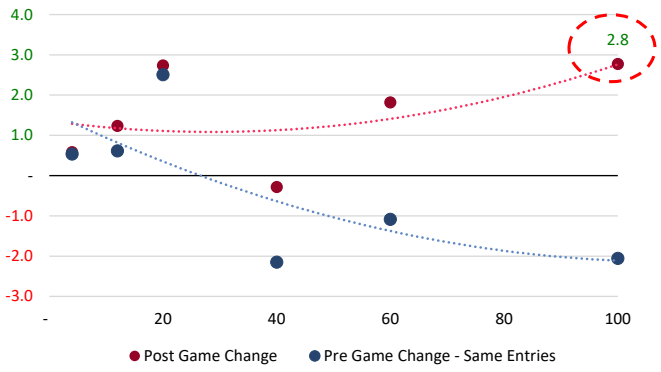
hypothetical sequence with the pre game change rules. If this sequence were pre game change, the sequence would have had a payout ratio above 60% (long-term legislated requirement) and a sequence deficit that would have depleted the prize reserve fund.

Exhibit 7 - Payout Ratio's (%, y-axis) based on Jackpot (A $m, x-axis) - sequence ending 3-Aug-23



Source: Company Data, E&P Estimates. Note: Includes WA

Exhibit 8 - Sequence Surplus/Defecit (A$m, yaxis) based on Jackpot (A$m, x-axis) - sequence ending 3-Aug-23



Source: Company Data, E&P Estimates. Note: Includes WA

Monte Carlo Simulation for FY24 vs prior years

Using our Monte Carlo simulation previously introduced, we have backtested prior years of Powerball to estimate the percentile the actual sales outcome was. The objective is to identify how much of an impact jackpot randomness had on a given result and determine the appropriate baseline for future projections.

The lines in the chart below represent the theoretical outcomes for each given year, with the actual result as the marker. In order to increase sales, TLC needs either one of two things;

1. Luck/Favourable Draws - this will move the marker up along the line

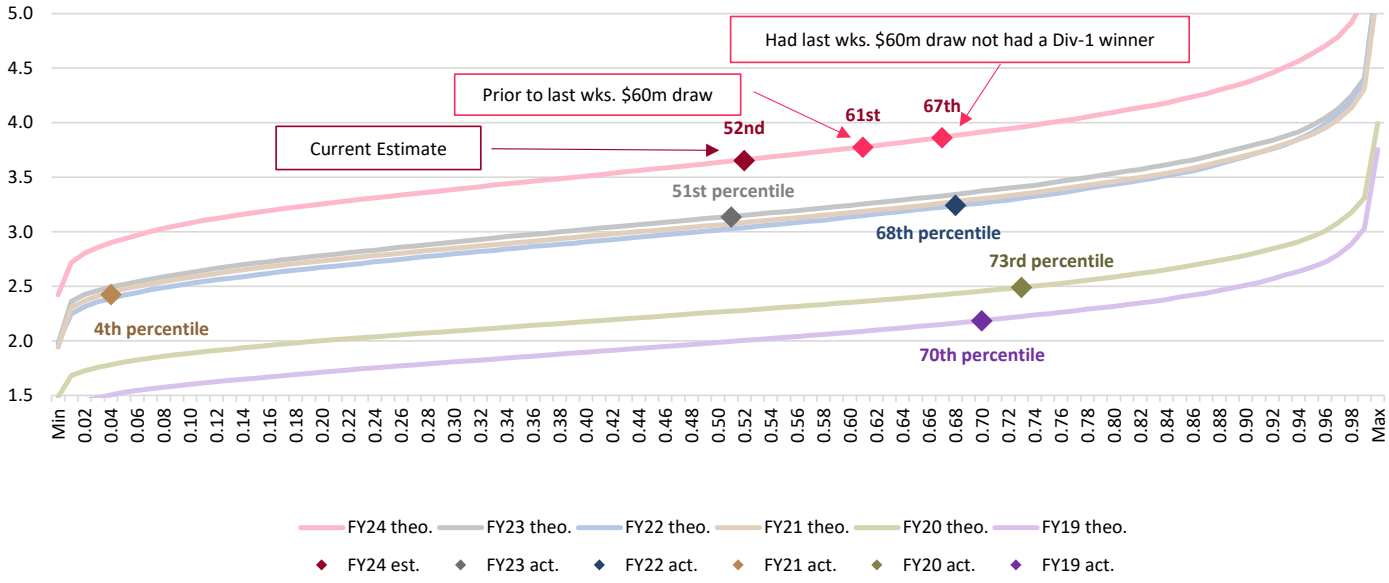
2. Improved simulation parameters - these will shift the entire line upwards.

Numerous factors go into determining these percentiles including: (i) weeks in period, (ii) starting jackpot, (iii) game changes, (iv) sequence impacts, (v) like-for-like entries, and (vi) the prize reserve fund. Factors such as the starting jackpot are short-term metrics, whilst like-for-like entries are more underlying factors reflective of the true strength of Powerball.

Below we can observe that the theoretical outcomes significantly increased post covid as the lines moved higher, before plateauing from FY21-FY23. We can also see that despite significant growth in the theoretical outcome in FY21, its sales declined -2.6% vs. FY20 due to a 4th percentile outcome. Similarly, FY23's decline vs. FY22 was a function of favorable jackpot outcomes in FY22, rather than a decline in theoretical outcomes.

We estimate FY24 to achieve record sales. This is largely due to the game change which has improved the theoretical potential outcomes. We estimate FY24 sales by adding the year-to-date sales with a 50th percentile outcome from a simulation for the remainder of the year. Prior to last week's $60m draw, Powerball was on track to generate a 61st percentile year. Had the $60m division-one jackpot not been won and the jackpot increased to $100m, Powerball would have been on track for a 67th percentile year. However, because there was a division-one winner, Powerball is now projected to have a 52nd percentile outcome with -5.4% lower sales than the alternative.

Exhibit 9 - Powerball Monte Carlo simulation backtesting & FY24 forecast - Sales (A$m, y-axis), percentile outcome (#, x-axis).



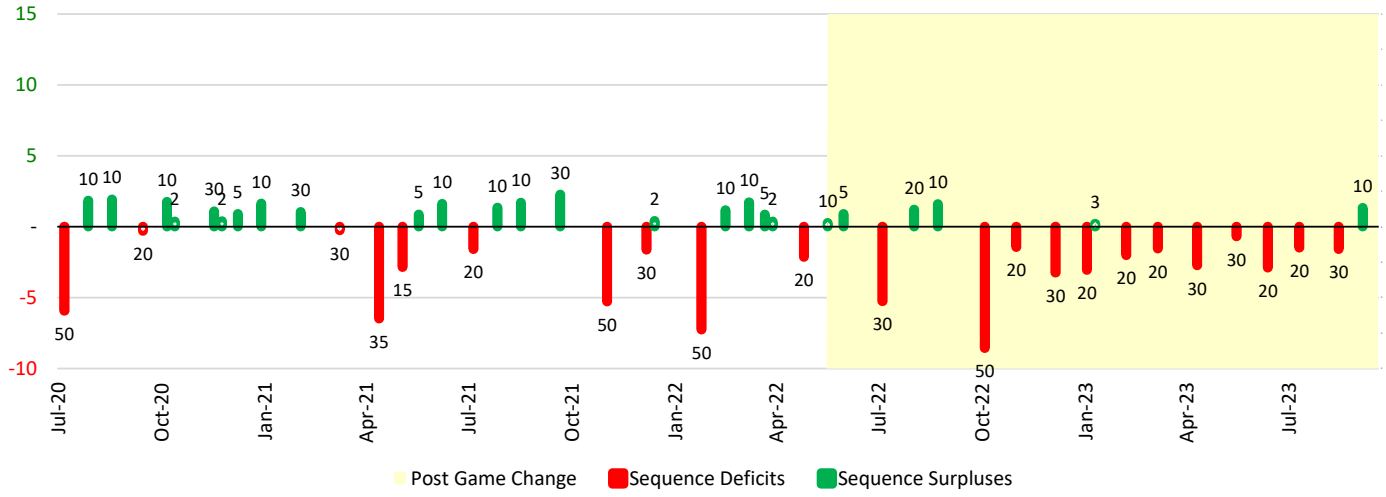
Source: Company Data, E&P Estimates. Note Includes WA.

Oz Lotto - Running low on Gas (prize reserve fund)

Prize Reserve Fund Analysis

In contrast to recent Powerball weeks, Oz Lotto has been depleting its prize reserve fund over the past year, a key variable that we flagged at the FY23 result. This has continued in FY24 YTD generally up until the most recent sequence.

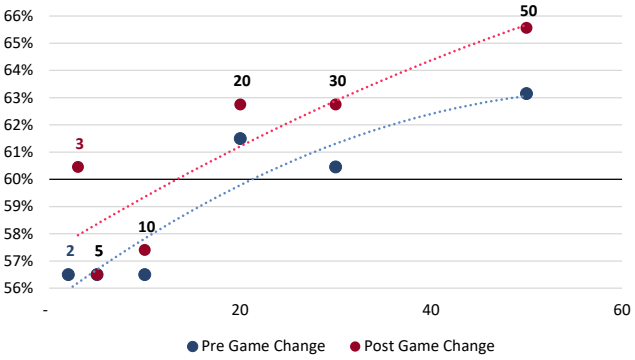
Exhibit 10 - Oz Lotto Net Sequence Prize Reserve Fund Accumulation (A$m) - jackpot as label (A$m)



Source: Company Data, E&P Estimates. Note Includes WA.

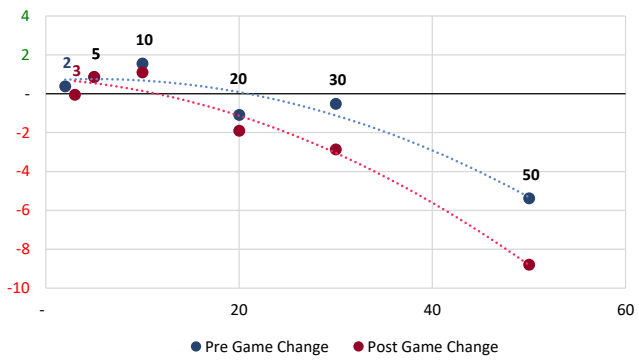
The May-22 game change resulted in an increase in the payout ratios and therefore a decrease in net prize fund accumulation despite a very similar sequence (only replaced $2m with $3m in week one). This is the opposite effect the Powerball game change had on the prize reserve fund.

Exhibit 11 - Theoretical relationship between Oz Lotto winning Div-1 Jackpot (A$m, x-axis) and Cumulative Sequence Payout ratio (%, y-axis)



Source: Company Data, E&P Estimates. Note: Includes WA. Label indicates Jackpot (A$m)

Exhibit 12 - Theoretical relationship between winning Div-1 Jackpot (A$m, x-axis) and Net Prize Fund Accumulation (A $m, y-axis)



Source: Company Data, E&P Estimates. Note: Includes WA. Label indicates Jackpot (A$m)

The below table outlines the recent game changes and key factors associated. Both the recent Powerball & Oz Lotto changes had a $10c increase in subscription price. However, the Oz Lotto game change also included a lengthening of the division-one odds and higher payouts to non-division-one.

Exhibit 13 - Summary of Recent Game Changes

	Powerball	Set for life	Saturday Lotto	Oz Lotto	Powerball	Friday Lotto
Date	**19-Apr-18**	**23-Mar-20**	**10-Oct-20**	**17-May-22**	**25-May-23**	**TBD**
Game changes						
Odds (Div. 1)	Longer	-	-	Longer	-	Extention of
Odds (Any prize)	Shorter	Shorter	Shorter	Shorter	-	M&W Lotto
Additional division	Div 9	Div 2 (1)	-	-	-	
Price increase	Yes	Yes	Yes	Yes	Yes	
Subscription Price						
Old	$0.85	$0.55	$0.65	$1.20	$1.10	
New	$1.10	$0.60	$0.75	$1.30	$1.20	
Increase	29.4%	9.1%	15.4%	8.3%	9.1%	
Combinations						
Old	76,767,600	-	-	45,379,620	134,490,400	
New	134,490,400	-	-	62,891,499	134,490,400	
Increase	75.2%	-	-	38.6%	0.0%	

Source: Company Data, E&P Estimates

The reason Oz Lotto & Powerball's game change had a differing affect on the prize reserve fund is a decline in Oz Lotto like-for-like entries which can be observed at each jackpot size. Although both Powerball & Oz Lotto had the same 10c price increase, we suspect Oz Lotto entries have particularly been impacted because of the following reasons:

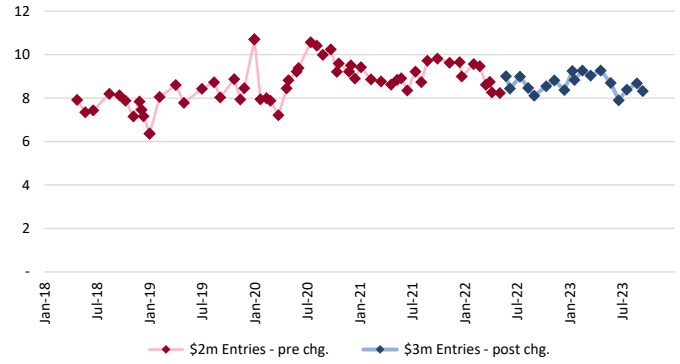
1. Lengthening of division-1 odds. This has the theoretical benefits of generating more large jackpots but reduces the return to player at each like-for-like jackpot level.

2. Lack of large Oz Lotto jackpots. Oz Lotto has only had one $50m+ jackpot since the May-22 game change (despite much higher theoretical jackpots). This has the potential to reduce the ability to attract new players and will reduce the amount of marketing the game receives.

3. Powerball strength. Powerball & Oz Lotto had very similar turnover from FY08 to FY18 with each having periods of being the largest jackpot game. Since the Apr-18 Powerball game change, the debate as to which is the dominant game has been settled. We view lotteries spending not just for individual games but as a whole portfolio and therefore believe that Powerball's growth has impacted recent Oz Lotto entries.

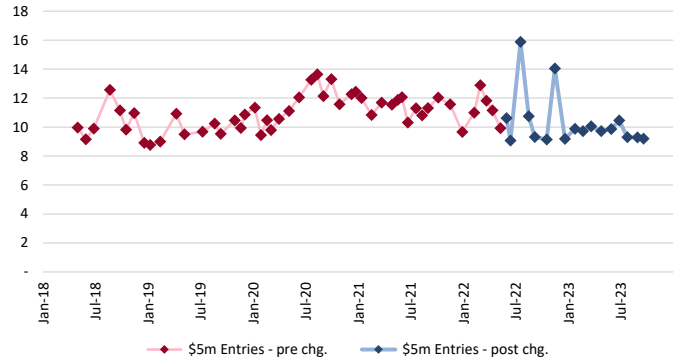
4. Cost of living pressures. Later in this note, we explore a recent decline in consumer spending. Oz Lotto is not the only game with flat or declining like-for-like growth, with most games' ex Powerball stagnating post lockdowns.

Exhibit 14 - OZL Entries (m's) when Jackpot is $2/3m



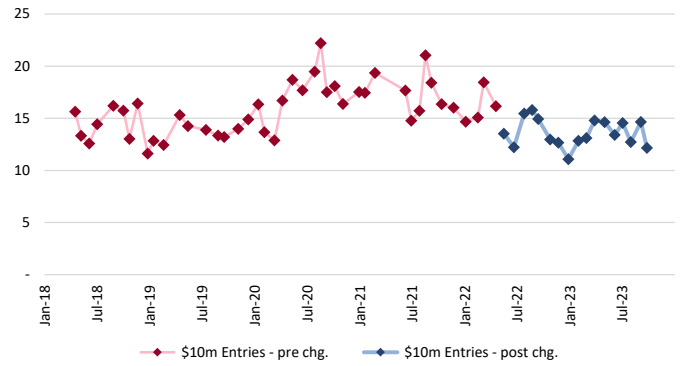
Source: Company Data, E&P Estimates. Note: Includes WA

Exhibit 15 - OZL Entries (m's) when Jackpot is $5m



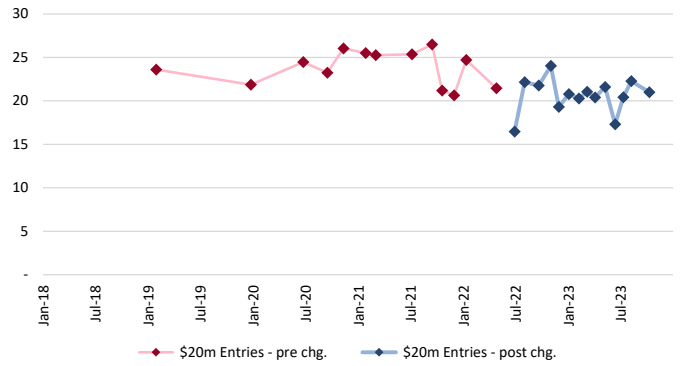
Source: Company Data, E&P Estimates. Note: Includes WA

Exhibit 16 - OZL Entries (m's) when Jackpot is $10m



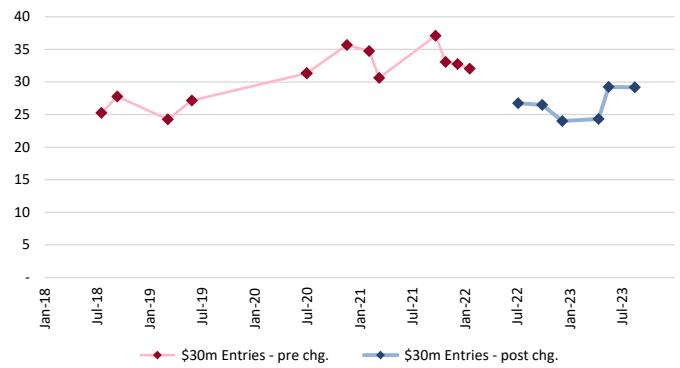
Source: Company Data, E&P Estimates. Note: Includes WA

Exhibit 17 - OZL Entries (m's) when Jackpot is $20m



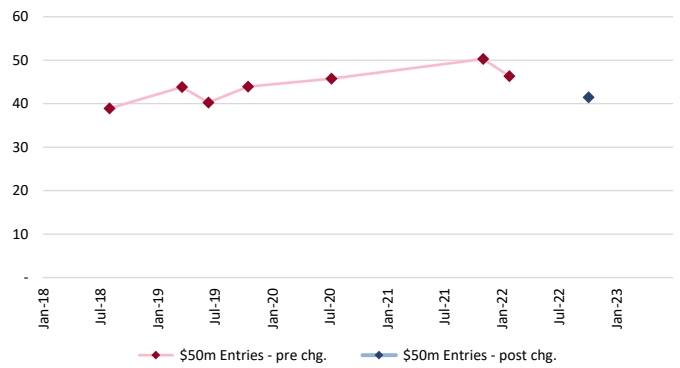
Source: Company Data, E&P Estimates. Note: Includes WA

Exhibit 18 - OZL Entries (m's) when Jackpot is $30m



Source: Company Data, E&P Estimates. Note: Includes WA

Exhibit 19 - OZL Entries (m's) when Jackpot is $50m

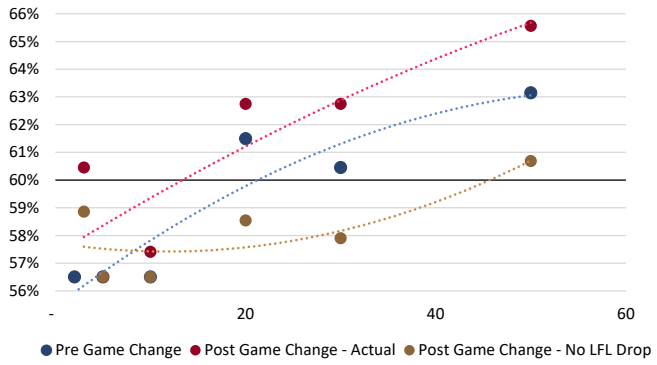


Source: Company Data, E&P Estimates. Note: Includes WA

If entries hadn't declined post game change, there would not be any concern about a depleting prize reserve fund. The payout ratios would have declined and the sequence surpluses would have improved from the blue lines to the brown lines. However, the decrease in like-for-like entries has more than offset the gains from raising the subscription price and consequently Oz Lotto has moved towards

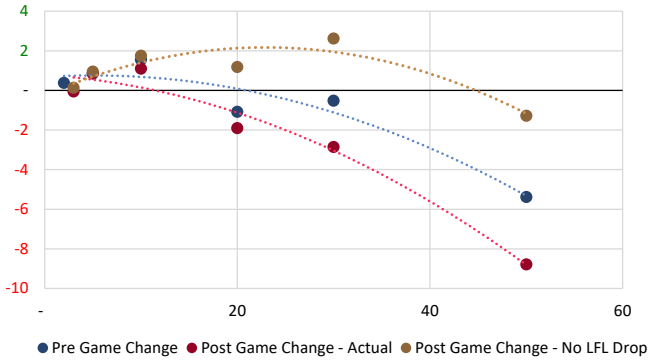
the red lines instead (i.e. payout ratios generally above 60%). Given the post game change payout ratios are mostly above the long term legislated requirement, the sequences are experiencing prize reserve fund deficits and are not self-sustaining.

Exhibit 20 - Theoretical relationship between Oz Lotto winning Div-1 Jackpot (A$m, x-axis) and Cumulative Sequence Payout ratio (%, yaxis) - game change comparison



Source: Company Data, E&P Estimates. Note: Includes WA

Exhibit 21 - Theoretical relationship between winning Div-1 Jackpot (A$m, x-axis) and Net Prize Fund Accumulation (A $m, y-axis) - game change comparison

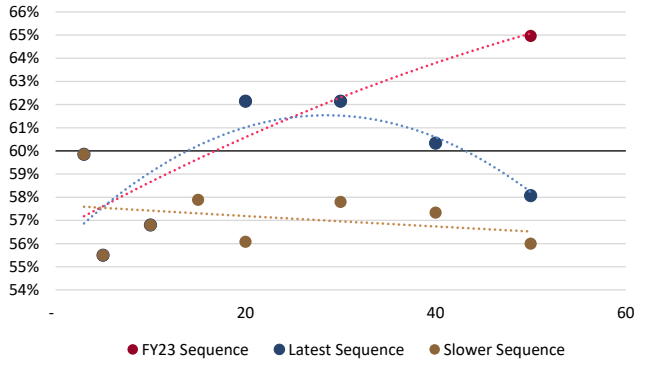


Source: Company Data, E&P Estimates. Note: Includes WA

Because the sequence used in FY23 is unsustainable over the long term with the current like-for-like entries, management needed to slow the sequence down.

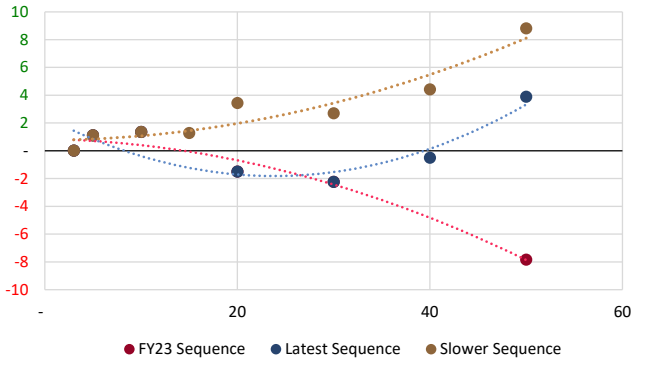
We believe they had already added a $40m jackpot into the latest sequence, pushing the $50m jackpot out to week seven (this wasn't observed because the jackpot didn't get high enough). This would make a $50m jackpot far more sustainable. Recently management added an additional $15m jackpot into week four. With this addition, Oz Lotto is expected to have a sequence surplus at every jackpot outcome, presuming like-for-like entries remain stable.

Exhibit 22 - Theoretical relationship between Oz Lotto winning Div-1 Jackpot (A$m, x-axis) and Cumulative Sequence Payout ratio (%, yaxis) - sequence comparision



Source: Company Data, E&P Estimates. Note: Includes WA

Exhibit 23 - Theoretical relationship between winning Div-1 Jackpot (A$m, x-axis) and Net Prize Fund Accumulation (A $m, y-axis) - sequence comparision



Source: Company Data, E&P Estimates. Note: Includes WA

Below we have determined the viability of each sequence. We have multiplied the theoretical net sequence prize reserve fund accumulation by the probability of each potential jackpot being the winning div-1 jackpot. If we sum these probability-weighted values (far right column), we can determine whether the sequence is expected to appreciate or reduce the prize reserve fund over the long term.

Based on the average entries since the game change, the baseline FY23 sequence would deplete the prize reserve fund while the slower alternative would increase the fund. The latest sequence has a marginally positive increase which begs the question as to why management needed to slow the sequence even further. The answer is that even though the latest sequence is expected to accumulate over the long-run, the randomness of the jackpots could easily cause depletion. This exact scenario has materialized as ten of the last twelve division-one winning jackpots were $20m or $30m, both of which would typically deplete the prize reserve fund under the latest sequence. We have previously noted that it is almost impossible to consistently have a sequence with zero impact on the prize reserve fund over a long period of time and that acceleration and deceleration are both natural and necessary.

An additional concern is that like-for-like entries could decline even further. We believe that given the significant depletion over the last twelve months, management will continue with the slower sequence, as they need to ensure Oz Lotto builds up the prize reserve fund again. The potential downside of the slower sequence is that it has several jackpots that are close to requiring 'top ups' to the division-one prizes, which is an unideal situation (see Burger with the Lott).

We note that we are unsure what the sequence looks like past the $50m jackpot and what the corresponding entries would be as this has not been achieved since Jun-19. Therefore, the sum of the winning div-1 jackpot probabilities in the column second from the right does not equal one.

Exhibit 24 - Theoretical Oz Lotto winning Sequence Payout ratio and Net Sequence Prize Reserve Fund Accumulation

Net Sequence Prize Reserve Fund Accumulation based on Speed of Sequence								
Week in Sequence	Jackpot	Entries	Payout Ratio	Net Sequence PRF Accumulation	Prob. of J/Pot going off	Prob. of reaching J/Pot	Prob. of being the Sequence J/Pot	Probability Weighted Net Sequence PRF Accumulation
FY23 SEQUENCE								
1	3,000,000	8,690,162	59.9%	16,355	12.9%	100.0%	12.9%	2,111
2	5,000,000	10,325,931	55.5%	1,112,441	15.1%	87.1%	13.2%	146,703
3	10,000,000	13,708,491	56.8%	1,358,703	19.6%	73.9%	14.5%	196,669
4	20,000,000	20,605,115	62.1%	-1,489,261	27.9%	59.4%	16.6%	-247,271
5	30,000,000	26,662,706	62.1%	-2,234,636	34.6%	42.8%	14.8%	-330,706
6	50,000,000	41,464,496	65.0%	-7,842,310	48.3%	28.0%	13.5%	-1,061,246
							SUM	**-1,293,740**
LATEST SEQUENCE								
1	3,000,000	8,690,162	59.9%	16,355	12.9%	100.0%	12.9%	2,111
2	5,000,000	10,325,931	55.5%	1,112,441	15.1%	87.1%	13.2%	146,703
3	10,000,000	13,708,491	56.8%	1,358,703	19.6%	73.9%	14.5%	196,669
4	20,000,000	20,605,115	62.1%	-1,489,261	27.9%	59.4%	16.6%	-247,271
5	30,000,000	26,662,706	62.1%	-2,234,636	34.6%	42.8%	14.8%	-330,706
6	40,000,000	33,805,500	60.3%	-500,747	41.6%	28.0%	11.7%	-58,361
7	50,000,000	41,464,496	58.1%	3,891,579	48.3%	16.4%	7.9%	307,649
							SUM	**16,794**
SLOWER SEQUENCE								
1	3,000,000	8,690,162	59.9%	16,355	12.9%	100.0%	12.9%	2,111
2	5,000,000	10,325,931	55.5%	1,112,441	15.1%	87.1%	13.2%	146,703
3	10,000,000	13,708,491	56.8%	1,358,703	19.6%	73.9%	14.5%	196,669
4	15,000,000	14,203,781	57.9%	1,288,835	20.2%	59.4%	12.0%	154,850
5	20,000,000	20,605,115	56.1%	3,440,871	27.9%	47.4%	13.2%	455,813
6	30,000,000	26,662,706	57.8%	2,695,496	34.6%	34.2%	11.8%	318,266
7	40,000,000	33,805,500	57.3%	4,429,385	41.6%	22.4%	9.3%	411,875
8	50,000,000	41,464,496	56.0%	8,821,712	48.3%	13.1%	6.3%	556,415
							SUM	**2,242,702**

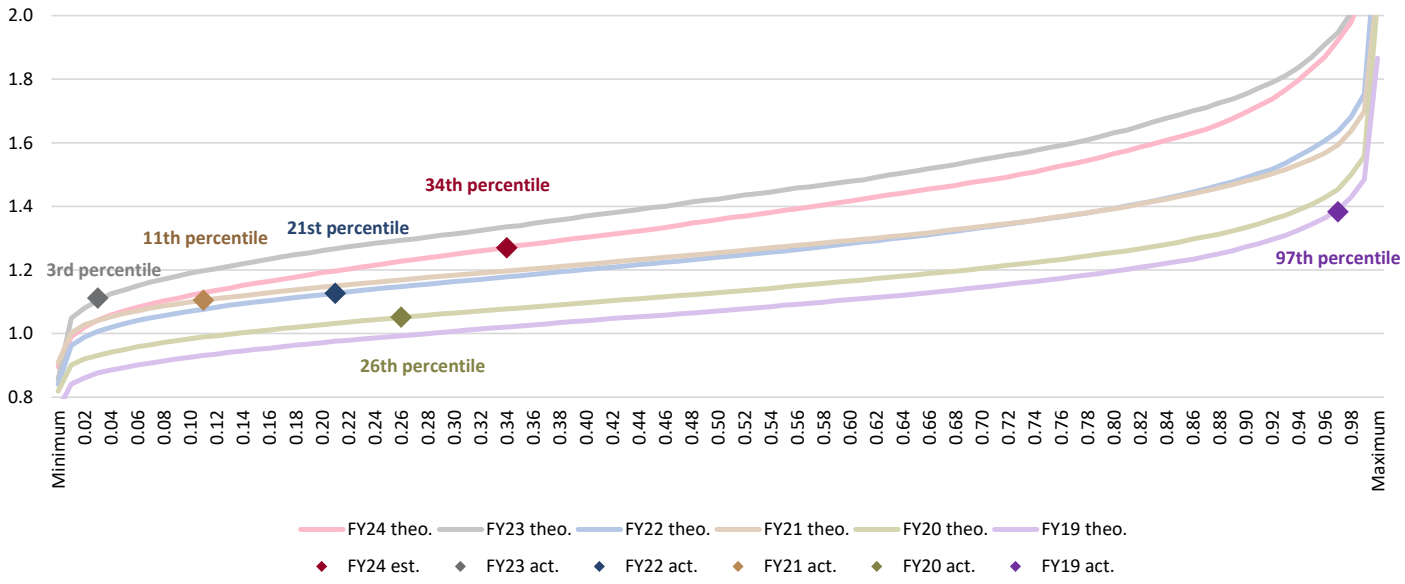
Source: Company Data, E&P Estimates. Note: Includes WA

Monte Carlo Simulation for FY24 vs prior years

Below is the same Monte Carlo backtesting chart explained above in the Powerball section but for Oz Lotto. The first thing you may notice is that Oz Lotto has had four years in a row that were the 26th percentile all less. This may seem unlikely but if you look at the history of the jackpots, it has not been able to exceed $50m once during that four-year stretch. This differs from FY19 which alone had four jackpots above $50m despite having a lower probability of doing so than each of the following years.

As described in the section above, we expect Oz Lotto to adopt slower sequences vs. FY23 and therefore expect FY24 to have lower theoretical outcomes. However, FY23 was extremely unlucky (see Burger with the Lott) and in the 3rd percentile. We anticipate it would take a 7th percentile run in FY24 to match FY23 sales. Thus far in FY24, Oz Lotto has had another unfavorable start. However, the year-to-date sales plus 50th percentile simulation for the remainder of the year would result in a 34th percentile result overall, which has +14.2% revenue upside vs. FY23.

Exhibit 25 - Oz Lotto monte carlo simulation backtesting & FY24 forecast - Sales (A$m, y-axis), percentile outcome (#, x-axis).



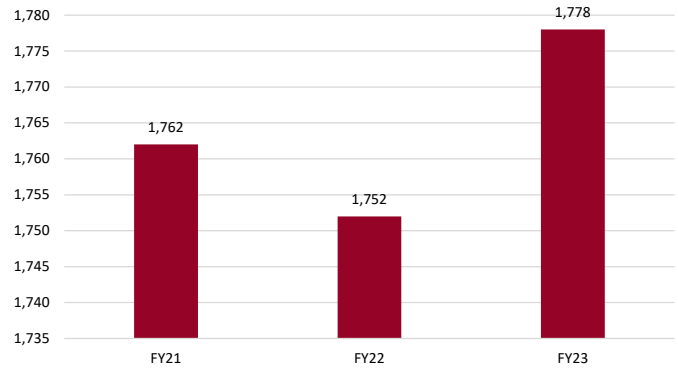
Source: Company Data, E&P Estimates. Note: Includes WA

Saturday Lotto - Smoothing Turnover

Turnover

Saturday Lotto achieved a record-high turnover in FY23, surpassing the previous record set in FY21 by +0.9%.

Exhibit 26 - Saturday Lotto Turnover (A$m)



Source: Company Data. Note: Ex-WA

Like-for-Like Entries

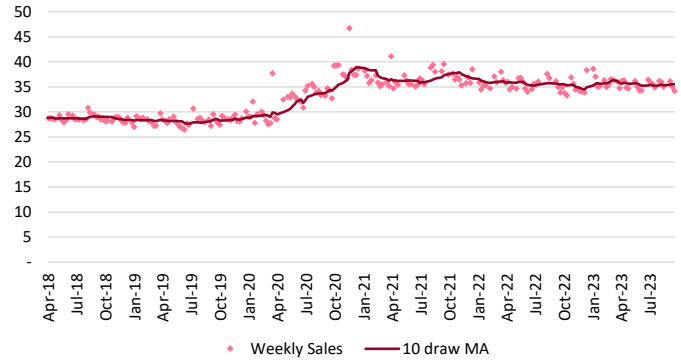
However, when we look at average like-for-like entries across the prior three years, we can observe FY23 to be inferior to FY21 at all three frequent jackpots.

Exhibit 27 - Saturday Lotto Like-for-Like Jackpot Entries

SATURDAY LOTTO ENTRIES								
Jackpot	Frequency			Avg. Entries			FY23 vs.	
	FY21	FY22	FY23	FY21	FY22	FY23	FY21	FY22
Normal	42	38	38	45,693,763	43,977,790	42,809,993	-6.3%	-2.7%
10,000,000	2	6	4	61,518,790	57,239,682	56,059,826	-8.9%	-2.1%
20,000,000	7	7	6	95,314,606	82,535,609	87,714,694	-8.0%	6.3%

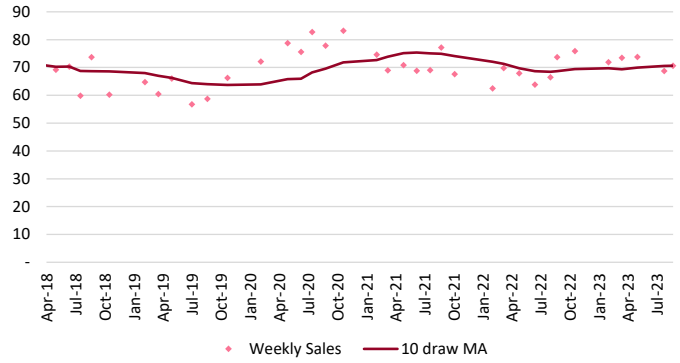
Source: Company Data, E&P Estimates. Note: Includes WA

Exhibit 28 - SL Sales (m's) for normal draws



Source: Company Data, E&P Estimates. Note: Includes WA

Exhibit 29 - SL Sales (m's) for $20m Superdraws



Source: Company Data, E&P Estimates. Note: Includes WA

Prize Reserve Fund

While the relationship may appear counterintuitive at first, the answer lies in the status of the prize reserve fund. The table below shows that FY21 significantly increased the prize reserve fund whilst FY23 depleted it. We do note there is some minor impact of the game change, but this increase/decrease is primarily a function of the number of Super/Megadraws that management opts to use.

Exhibit 30 - Saturday Lotto Net Prize Reserve Fund Accumulation (A$m)

	Subsriptions	Prizes	Theoretical Prizes	Net PRF Accumulation
FY21	2,083,911,853	1,219,068,144	1,250,347,112	31,278,968
FY22	2,080,843,777	1,247,065,858	1,248,506,266	1,440,408
FY23	2,119,361,520	1,286,435,398	1,271,616,912	-14,818,486

Source: Company Data, E&P Estimates. Note: Includes WA

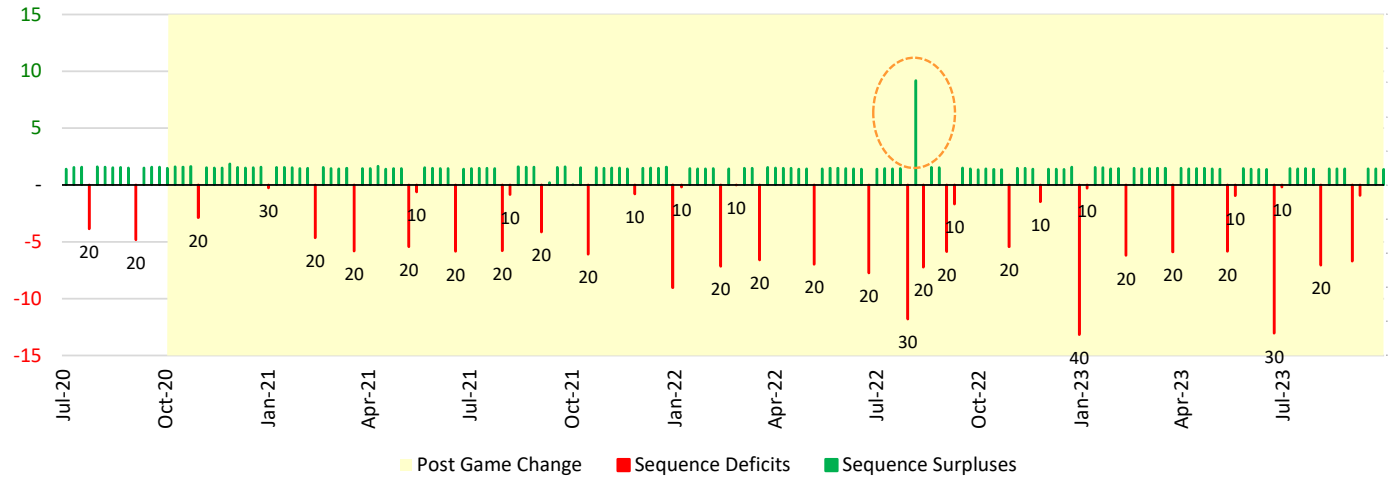
The accumulation and depletion of the prize reserve fund is much simpler for Saturday Lotto than it is for Powerball & Oz Lotto. Rather than being based on an entire sequence, it is on a week-to-week basis.

Whenever a normal draw occurs, the division-one prize will be approximately 16.65% of weekly subscriptions. This is measured as the minimum prize payout per each individual draw (currently 55.5%) multiplied by the allocation towards division-one (currently 30%). The total long-term prize payout is 60% of subscriptions (same as Powerball & Oz Lotto) with the remaining 4.5% of subscriptions used to fill up the prize reserve fund. Super/Megadraws use funds from the prize reserve fund to top up the division-one prize above the minimum legislated amount. Theoretically, they can still result in a weekly prize reserve fund surplus if the subscriptions are considerably large enough but usually result in a weekly prize fund deficit.

In each instance, divisions two-to-six will receive a predetermined proportion of the subscriptions (70% x 55.5% across the five divisions).

A major outlier occurs if there are no division-one winners. In these instances, the prize allocation for division-one is supposed to be added to the next draw. However, if the next draw is a Superdraw/Megadraw, then TLC would be paying above the weekly minimum prize anyway, resulting in the week without a division-one winner to essentially become an incredibly favorable outcome that will boost the prize reserve fund.

Exhibit 31 - Saturday Lotto Net Weekly Prize Reserve Fund Accumulation (A$m) - Super/Megadraw Jackpots as labels (A $m)



Source: Company Data, E&P Estimates. Note: Includes WA

For FY23, the breakdown of weekly surplus/deficit by jackpot is listed below. As expected, the greater the division-one prize, the greater the number of subscriptions, but also the faster the prize reserve fund will deplete. There are a few other key things to note that are

specific to FY23. The first is that there were three Megadraws ($30/40m). Historically there has always been one on New Year's Eve, with a mid-year (usually EOFY) draw occurring sporadically. In FY23, the mid-year draws happened in Jul-22 and Jun-23, which both happened to fall in the same reporting period.

The other key observation is that the no-winner draw should be viewed as a significant outlier. In this instance, it occurred when a $10m division-one prize was offered. This is extremely favorable for TLC but also extremely unlikely. The odds of winning Saturday Lotto are 1 in 8,145,060 and that particular draw had ~57.9m entries. Thus, the probability of no winners was 0.08% or 1 in every 1,223 draws. The odds of this happening for a normal draw is one in every 221 draws. Given this, we believe the prize reserve fund depletion in FY23 would be even greater under a normalized year.

We believe that management opted to use an aggressive sequence to offset the weaker year observed in FY23 overall vs. FY22. This is perfectly rational and offers TLC some level of flexibility in smoothing their turnover to account for the randomness seen by Powerball and Oz Lotto. However, they are only able to do this over the short term if they have previously built up sufficient prize reserves, such as in FY21 (a strong year overall), or are willing to use a slow sequence in the near future.

Exhibit 32 - FY23 Saturday Lotto Net Prize Reserve Fund Accumulation (A$m) by Jackpot

FY23 Saturday Lotto Subscriptions & PRF					
D1 Prize	Count	Wk. Subs	Wk. Net PRF	Cum. Subs	Cum. Net PRF
No Winner	1	43,428,001	9,185,022	43,428,001	9,185,022
Normal Draw	38	32,107,495	1,446,960	1,220,084,805	54,984,489
10,000,000	4	42,044,870	(1,107,510)	168,179,478	(4,430,040)
20,000,000	6	65,786,020	(6,086,257)	394,716,121	(36,517,540)
30,000,000	2	83,071,488	(12,430,380)	166,142,976	(24,860,761)
40,000,000	1	126,810,139	(13,179,656)	126,810,139	(13,179,656)
	52			**2,119,361,520**	**(14,818,486)**

Source: Company Data, E&P Estimates. Note: Includes WA

Determining the Ideal Sequence

Over the long term, management should aim to maximize subscriptions, with close to zero impact on the prize reserve fund. The table below demonstrates how many normal draws it would take to offset the prize reserve deficit of a Super/Megadraw. We have split the Megadraws up individually because we have identified that NYE draws are superior to mid-year draws. We also note that despite the $30m Jun-23 and $40m NYE-22 draws appearing to have weaker subscriptions than their prior instances, this is potentially because they both followed $100m Powerball draws and likely suffered from cannibalisation.

Exhibit 33 - Number of Normal draws required to offset a Super/Megadraw

Saturday Lotto PRF Accumulation by Jackpot			
Jackpot	Wk. Subs	Wk. Net PRF	Normal Draws to offset
Frequent Draws (avg.)			
Normal	32,803,960	1,477,868	
10m	43,379,490	(443,831)	0.30x
20m	64,223,681	(5,971,900)	4.04x
One-off Megadraws			
30m Jun '23	80,150,517	(13,048,166)	8.83x
40m NYE '22	126,810,139	(13,179,656)	8.92x
30m Jul '22	85,992,458	(11,812,595)	7.99x
40m NYE '21	136,586,233	(9,027,364)	6.11x
30m NYE '20	131,115,025	(268,029)	0.18x

Source: Company Data, E&P Estimates. Note: Includes WA. Average subscriptions and net PRF accumulation based on trailing 12-months

Below we use goal-seek to obtain a prize-fund-neutral year to determine which jackpots offer could potentially maximise subscriptions. We first observe that the $40m NYE Megadraw appears to be the best. However, in contrast, the $30m mid-year Megadraws yields the least subscriptions. The other main thing we observe is that $10m Superdraws are superior to $20m Superdraws. Management has potentially recognized this as $10m draws were introduced in FY21 and appear to have become a staple jackpot.

If the below analysis were the only factors to consider, then we would recommend using one NYE $40m draw, and exclusively $10m draws in conjunction with normal draws. However, we are unaware of how much marketing can be attributable to each jackpot. We also speculate that the $20m jackpots are considered somewhat of a brand drawcard, and that players may be potentially turned away from the game if they are removed permanently.

Exhibit 34 - Saturday Lotto Jackpot scenarios that are Prize Reserve Fund Neutral

Saturday Lotto Jackpot Frequencies - Prize Fund Neutral							
Jackpot	Wk. Net PRF	Weeks	Total PRF	Wk. Subs	Total. Subs	Total. Subs ex-WA	vs. FY23A
$10m Superdraw							
Normal	1,438,761	18.14	26,100,834	31,940,529	579,439,213		
10m	(770,873)	33.86	(26,100,834)	43,636,535	1,477,481,137		
		52.00	-		2,056,920,350	1,725,501,571	-2.95%
$20m Superdraw							
Normal	1,438,761	42.20	60,711,871	31,940,529	1,347,805,171		
20m	(6,193,404)	9.80	(60,711,871)	65,279,412	639,912,246		
		52.00	-		1,987,717,416	1,667,448,875	-6.22%
$30m - June 2023							
Normal	1,438,761	46.84	67,385,297	31,940,529	1,495,955,412		
30m Jun '23	(13,048,166)	5.16	(67,385,297)	80,150,517	413,925,345		
		52.00	-		1,909,880,757	1,602,153,553	-9.89%
$30m - July 2022							
Normal	1,438,761	46.35	66,692,491	31,940,529	1,480,575,105		
30m Jul '22	(11,812,595)	5.65	(66,692,491)	85,992,458	485,503,079		
		52.00	-		1,966,078,184	1,649,296,238	-7.24%
$40m - NYE 2022							
Normal	1,438,761	46.88	67,452,131	31,940,529	1,497,439,128		
40m NYE '22	(13,179,656)	5.12	(67,452,131)	126,810,139	649,001,331		
		52.00	-		2,146,440,459	1,800,597,861	+1.27%
$40m - NYE 2021							
Normal	1,438,761	44.85	64,530,791	31,940,529	1,432,585,302		
40m NYE '21	(9,027,364)	7.15	(64,530,791)	136,586,233	976,366,662		
		52.00	-		2,408,951,964	2,020,812,521	+13.66%

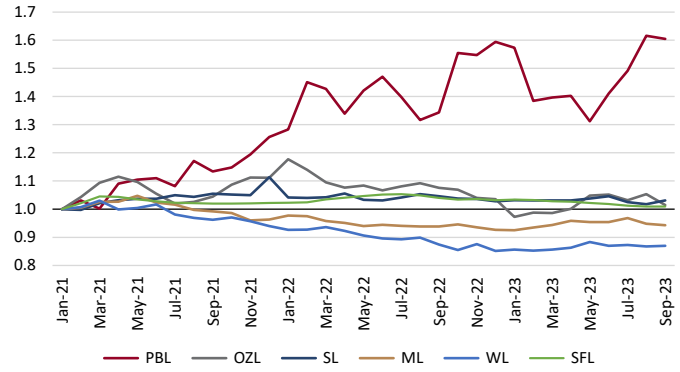
Source: Company Data, E&P Estimates. Note: Includes WA. Average subscriptions and net PRF accumulation based on trailing 12-months

Overall Portfolio

Below we demonstrate the performance of each game post Covid. We use the twelve-month moving average total to smooth out the sporadic nature of lotteries turnover which is often skewed by large jackpots. Powerball is the clear outlier and continues going from strength to strength. The other games are relatively flat or slightly declining, albeit we note that there was strong growth during the lockdowns that this data is in effect comping.

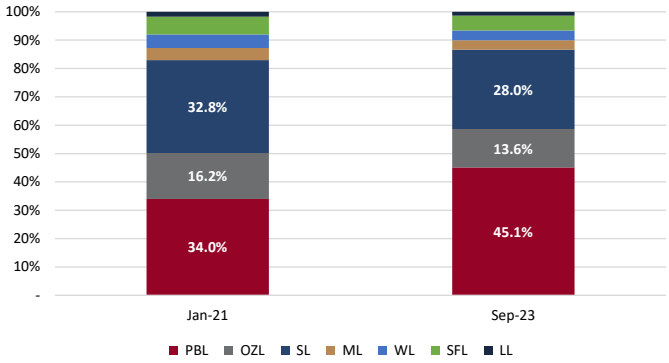
The result is an increasing shift in dependence on Powerball, increasing from 34.0% of the portfolio to 45.1% (excl. instants & other games). Given Powerball is a jackpot game whose performance is dependent on randomness, TLC earnings have become increasingly volatile. This increases the importance of understanding management's ability to accelerate or slowdown their jackpot sequence, subject to the prize reserve fund, to smooth out overall turnover.

Exhibit 35 - Lotteries Turnover - 12 month Moving Average Total index (Jan-21 = 1.0)



Source: Company Data, E&P Estimates

Exhibit 36 - Proportion of Lotteries Turnover per Game - 12 month Moving Average Total

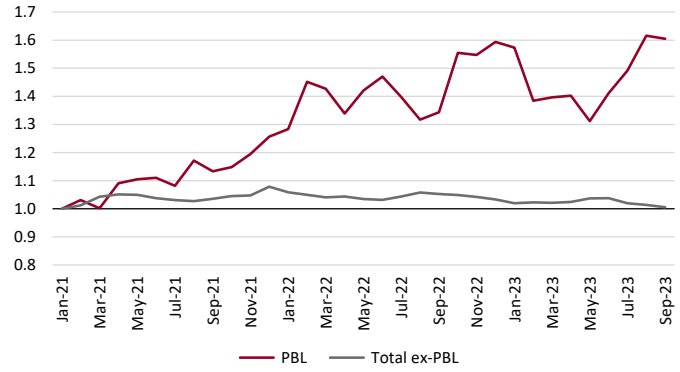


Source: Company Data, E&P Estimates. Note Excludes Instants and Other Games

Below we have charted Powerball vs. the rest of the portfolio. Since Jan-21, the twelve-month moving average for Powerball has increased by +60%, while the rest of the portfolio is up only +1%. We had previously stated that we believe cannibalisation exists between the games. Consequently, we suspect Powerball's growth has come at the expense of the other games.

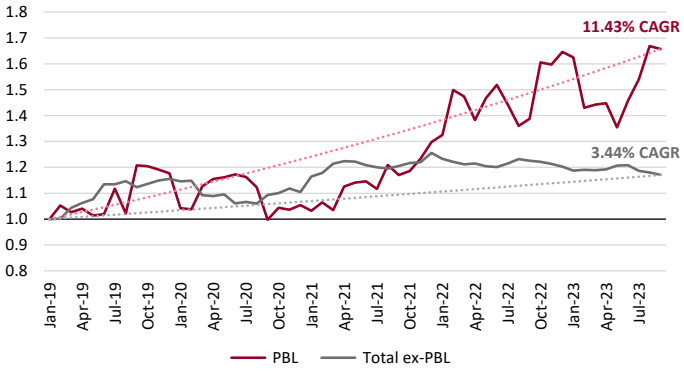
For context, we must also consider the abnormal nature of the lockdowns in conjuction with cannibalistion. If we factor this into consideration, the rest of the portfolio would have a respectable yearly CAGR of +3.44% despite being flat since Jan-21.

Exhibit 37 - Lotteries Turnover - 12 month Moving Average Total index (Jan-21 = 1.0)



Source: Company Data, E&P Estimates. Note Excludes Instants and Other Games

Lotteries Turnover - 12 month Moving Average Total index (Jan-19 = 1.0)



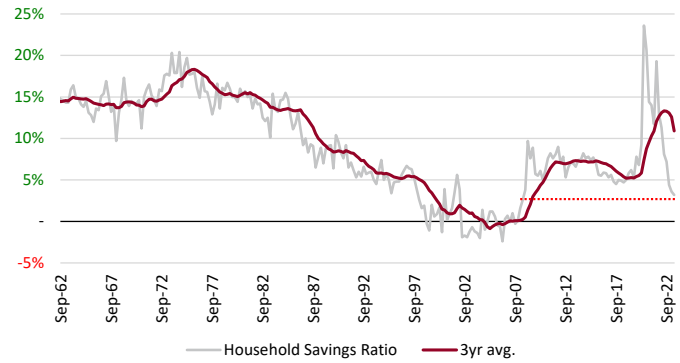
Source: Company Data, E&P Estimates. Note Excludes Instants and Other Games

Macro Analysis

The household savings ratio has continued to decline down to 3.2%, the lowest since Jun-08. Savings continue to fall nationwide as the increase in nominal household consumption outpaces the increase in gross disposable income. Household spending has also shifted in the past year, with non-discretionary spending yoy growth outpacing discretionary spending growth over the past 12-months. We believe these are both indicators of a rise in the cost of living pressures that have the potential to put downward pressure on

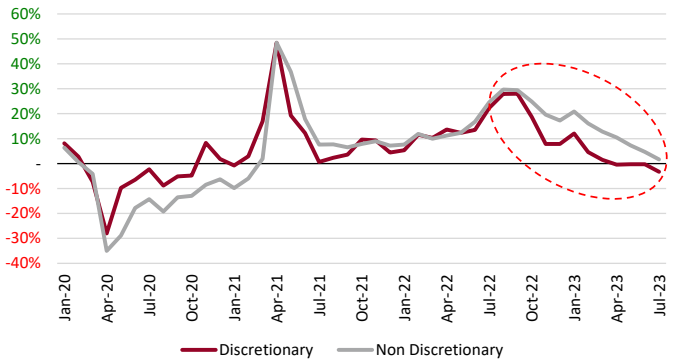
lottery sales. However, we do note that historically lotteries have shown greater resilience to economic slowdowns than other forms of gambling/entertainment.

Exhibit 38 - Household Savings Ratio



Source: ABS, E&P Estimates

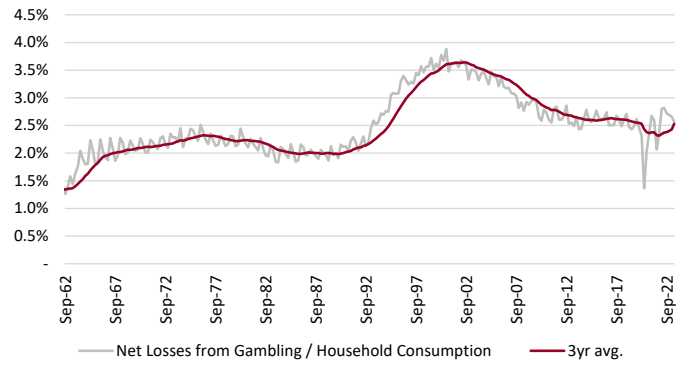
Exhibit 39 - Household Spend Index - yoy growth



Source: ABS, E&P Estimates

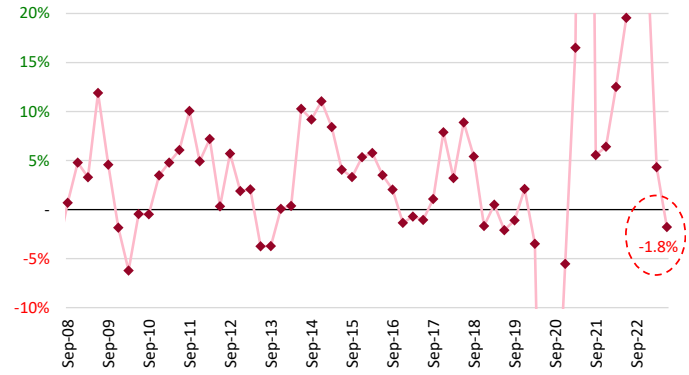
Overall gambling spend as a proportion of household income became elevated post-Covid lockdowns but appears to have stabilized recently. The recent June-23 quarter saw the first yoy decline since the pandemic.

Exhibit 40 - Gambling Spend as a proportion of Household Consumption



Source: ABS, E&P Estimates

Exhibit 41 - Household Gambling Expenditure - yoy growth



Source: ABS, E&P estimates

Earnings Changes

We had largely factored in the topics explored. Our changes to revenue come from adjustments to FY24 turnover which reflect the year-to-date performance of the portfolio of games. This involves marginal declines for most games offset, by an increase to Powerball to reflect the games' ability to maintain like-for-like entries post game change. We have also made minor increases to Opex and adjusted D&A to reflect guidance.

Exhibit 42 - TLC changes to financial forecasts

		FY24	FY25	FY26
Revenue	New	3785.4	3945.2	4086.0
	Old	3753.5	3945.2	4086.0
	% change	0.9%	0.0%	0.0%
EBITDA	New	779.8	819.3	856.2
	Old	799.0	848.2	886.2
	% change	-2.4%	-3.4%	-3.4%
EBIT	New	668.8	707.6	743.7
	Old	699.8	749.0	786.9
	% change	-4.4%	-5.5%	-5.5%
NPAT	New	377.0	400.0	425.4
	Old	398.8	429.2	456.3
	% change	-5.5%	-6.8%	-6.8%
EPS (cps)	New	16.9	18.0	19.1
	Old	17.9	19.3	20.5
	% change	-5.4%	-0.8%	-0.8%

Source: E&Pe

Our valuation declines marginally to $5.40/sh. We retain our Neutral recommendation albeit note that the upside implied in our valuation is decent at current levels. Our hesitation remains consensus expectations rather than the company itself.

We note that for all periods where data is not yet formed (ie. draws have not gone off) we model out a 50th percentile probable revenue outcome. As such our forecasts beyond 1H24 (or most of the time just the current period) will not have any controversial earnings calls in them presuming consensus is modelling similarly. For 1H24 our forecasts vs Visible Alpha consensus are as follows:

- Revenue $1876.6m vs VA consensus $1970.1m (-4.7%)
- EBITDA $388.3m vs VA consensus $407.9m (-4.8%)
- EBIT $332.9m vs VA consensus $351.1m (-5.2%)
- NPAT $187.5m vs VA consensus $195.9m (-4.3%)

As such at the moment we believe consensus is modelling a probability of a strong upward reversion in luck or like-for-like entries vs probable outcomes for the revenue line, which is driving higher forecasts for 1H24 than we expect. As such while we see decent valuation upside at the moment we remain Neutral.

The Lottery Corporation, Neutral, VALUATION AUD5.40

EAP CORE DRIVERS

- Household disposable income and population growth driving gambling spend.
- Product innovation in lotteries games.
- Online migration impacting revenue growth and margins.

INVESTMENT THESIS

- Sales growth driven by: (i) strong customer acquisition during Covid-19; and (ii) recent and future game changes.
- Higher margins as sales migrate online, with online sales EBITDA margins more than double that of retail.
- Capital management initiatives given significant FCF generation whilst being under the target gearing range.



FINANCIAL SUMMARY

The Lottery Corporation As at:	13/10/2023		TLC-ASX Recommendation:	Neutral	Share Price:	AUD4.81

Year end	Jun	2023A	2024E	2025E	2026E
INCOME STATEMENT					
Total Operating Income/Revenue		3,513.1	3,785.4	3,945.2	4,086.0
EBITDA		713.2	779.8	819.3	856.2
D&A		(98.2)	(111.0)	(111.7)	(112.5)
EBIT		615.0	668.8	707.6	743.7
Net Interest Expense		(121.9)	(130.3)	(136.1)	(136.0)
PBT		386.5	538.5	571.5	607.7
Tax expense		(121.7)	(161.6)	(171.4)	(182.3)
NPAT attributable to s'hers		**264.8**	**377.0**	**400.0**	**425.4**
Adjusted Net Profit		264.8	377.0	400.0	425.4
Year end	**Jun**	**2023A**	**2024E**	**2025E**	**2026E**
BALANCE SHEET					
Assets					
Current Receivables		41.8	49.8	51.6	53.4
Other Current Assets		450.1	450.1	450.1	450.1
Total Current Assets		926.4	1,129.4	1,237.2	1,368.6
Property Plant & Equipment		147	155	162	170
Intangibles		2,971.4	2,932.3	2,895.9	2,861.7
Other Non-Current Assets		275.4	275.4	275.4	275.4
Total Non-Current Assets		3,444.5	3,413.0	3,384.1	3,357.4
Total Assets (BS)		4,370.9	4,542.4	4,621.4	4,726.1
Liabilities					
Total Current liabilities		3,518.2	3,669.0	3,704.2	3,740.8
Total Non-Current Liabilities		587.9	587.9	587.9	587.9
Total Liabilities		4,106	4,257	4,292	4,329
Total Shareholders' Equity		264.8	285.5	329.3	398.4
Net Assets		264.8	285.5	329.3	397.4
Net Tangible Assets		(2,706.6)	(2,646.8)	(2,566.6)	(2,463.3)
Book Value		264.8	285.5	329.3	398.4
Year end	**Jun**	**2023A**	**2024E**	**2025E**	**2026E**
OPERATING CASH FLOWS					
EBITDA		704.6	779.8	819.3	856.2
Change in Working Capital		0.0	142.8	33.5	33.7
Gross Operating Cash Flow		**704.6**	**922.5**	**852.8**	**890.0**
Net interest paid		(117.2)	(130.3)	(136.1)	(136.0)
Tax Paid		(130.9)	(161.6)	(171.4)	(182.3)
Net Operating Cash Flows		**456.5**	**630.7**	**545.2**	**571.6**
Investing Cash Flows					
Purchase of PP&E		(79.6)	(79.5)	(82.8)	(85.8)
Other Investing Cashflows		(41.6)	0.0	0.0	0.0
Total Investing CashFlow		**(121.2)**	**(79.5)**	**(82.8)**	**(85.8)**

Source: Company data, E&P estimates

Year end	Jun	2023A	2024E	2025E	2026E
FINANCING CASH FLOWS					
Proceeds from Borrowings		0.0	0.0	0.0	0.0
Net Proceeds Issue/Buyback of Securities		(10.3)	0.0	0.0	0.0
Distributions		(197.1)	(356.3)	(356.3)	(356.3)
Total Financing Cash Flow		**(217.0)**	**(356.3)**	**(356.3)**	**(356.3)**
Net Change in Cash		118.3	194.9	106.1	129.6
Free Cashflow		376.9	551.2	462.4	485.8
Year end	**Jun**	**2023A**	**2024E**	**2025E**	**2026E**
PER SHARE DATA					
EPS - Adjusted (cents)		11.9	16.9	18.0	19.1
DPS (cents)		0.2	0.2	0.2	0.2
Cash Flow Per Share (cents/share)		16.9	24.8	20.8	21.8
Book Value ($/sh)		11.9	12.8	14.8	17.9
NTA Per Share ($/sh)		(121.6)	(118.9)	(115.3)	(110.6)
Year end	**Jun**	**2023A**	**2024E**	**2025E**	**2026E**
ANALYSIS					
EBITDA Margin (%)		20.3	20.6	20.8	21.0
EBIT Margin (%)		17.5	17.7	17.9	18.2
Net Profit Margin (%)		7.5	10.0	10.1	10.4
Capital Employed		2,290.1	2,115.9	2,053.5	1,992.1
Return on Capital Employed (%)		26.7	30.4	33.9	36.8
EBITDA/Capital Employed (%)		31.0	35.4	39.3	42.3
ROE %		**104.3**	**137.0**	**130.1**	**116.9**
Net Debt/Equity (%)		764.8	641.1	523.7	400.3
Net Debt/Net Debt & Equity (%)		88.4	86.5	84.0	80.0
Net Debt/EBITDA (x)		2.8	2.3	2.1	1.9
Net Interest Cover (x)		5.0	5.1	5.2	5.5
GOCF/EBITDA (%)		1.0	1.2	1.0	1.0
Total Capex/Sales (%)		2.3	2.1	2.1	2.1
Working Capital/Sales (%)		(0.2)	(0.3)	(0.3)	(0.3)
D&A/PP&E (%)		66.8	71.8	68.9	66.3
EV/EBITDA (x)		17.9	16.1	15.2	14.4

BLENDED VALUATION		
Valuation - DCF	5.40	100.0%
Valuation - Capitalisation of Earnings	0.00	0.0%
Valuation - PER	0.00	0.0%
Valuation - SOTP	0.00	
Valuation	AUD5.40	

RESEARCH RECOMMENDATION DEFINITIONS

Positive	Stock is expected to outperform the S&P/ASX 200 over the coming 24 months.
Neutral	Stock expected to perform in line with the S&P/ASX 200 over the coming 24 months.
Negative	Stock is expected to underperform the S&P/ASX 200 over the coming 24 months.
Speculative Buy	Stock has limited history from which to derive a fundamental investment view or its prospects are highly dependent on event risk, e.g. Successful exploration, scientific breakthrough, high commodity prices, regulatory change, etc. Consequently, the stock is considered a high-risk investment which may be prone to high volatility in share price movements, have a greater risk of capital loss and/or the stock may have low liquidity.
Suspended	Stock is temporarily suspended due to compliance with applicable regulatory and/or E&P policies in circumstances where E&P is acting in an advisory capacity.
Not Rated	Stock is not included in our investment research universe.

Research Criteria Definitions

Recommendations are primarily determined with reference to how a stock ranks relative to the S&P/ASX 200 on the following criteria:

Valuation	Composite of Rolling 12-month prospective multiples and discounted cash flow (DCF), or DCF for resource stocks.
Earnings Outlook	Forecast 2-year EPS growth.
Earnings Momentum	Percentage change in the current consensus EPS estimate for the stock (rolling 1-year forward basis) over the consensus EPS estimate for the stock 3 months ago.
Shareholder Returns	Composite of forecast ROE (rolling 1-year forward basis) and the percentage change in ROE over 2 years.
Debt Servicing Capacity	Rolling 12-month EBIT Interest Cover ratio.
Cyclical Risk	Qualitative assessment of the 2-year outlook for a stock/industry's profit cycle.
Industry Quality	Qualitative assessment of an industry's growth/returns potential and company specific management capability.
Financial Transparency	If we don't understand it, we won't recommend it.

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RESEARCH				
Cameron McDonald	Managing Director, Head of Research	Transport, Infra, Waste, Packaging	cameron.mcdonald@eandp.com.au	+61 3 9235 9709
Paul Mason	Managing Director	Technology	paul.mason@eandp.com.au	+61 3 9631 9867
Julian Mulcahy	Managing Director	Small Caps	julian.mulcahy@eandp.com.au	+61 3 9235 9713
Adam Martin	Executive Director	Energy	adam.martin@eandp.com.au	+61 3 9235 9724
Olivier Coulon	Executive Director	Small Caps	olivier.coulon@eandp.com.au	+61 2 8070 6634
Azib Khan	Executive Director	Banks	azib.khan@eandp.com.au	+61 2 8070 6636
Phillip Kimber	Executive Director	Consumer	phillip.kimber@eandp.com.au	+61 3 9631 9873
Robin Young	Executive Director	Wealth	robin.young@eandp.com.au	+61 3 9235 9730
Entcho Raykovski	Executive Director	Telco, Media	entcho.raykovski@eandp.com.au	+61 3 9235 9719
David Nayagam	Director	Healthcare	david.nayagam@eandp.com.au	+61 3 9235 9704
Harry Saunders	Associate Director	Building Materials	harry.saunders@eandp.com.au	+61 3 9235 9744
Johnny Huynh	Associate Director	Technology, Media, Telco	johnny.huynh@eandp.com.au	+61 3 9235 9737
Nick Weal	Associate Director	Small Caps	nick.weal@eandp.com.au	+61 2 80706637
Kieran Harris	Associate	Small Caps	kieran.harris@eandp.com.au	+61 3 9235 9749
Pallavi Malladi	Associate	Technology, Media, Telco	pallavi.malladi@eandp.com.au	+61 3 9631 9847
Kade Madigan	Associate	Consumer	kade.madigan@eandp.com.au	+61 2 8070 6637
Branko Skocic	Analyst	Energy	branko.skocic@eandp.com.au	+61 3 9235 9705
Alan Liang	Analyst	Healthcare	alan.liang@eandp.com.au	+61 3 92359771
SALES AND TRADING (AUSTRALIA)				
Alex Rock	Managing Director	Head of Institutional Equities	alex.rock@eandp.com.au	+61 2 8070 6617
Adam Woods	Managing Director	Head of Sales Trading	adam.woods@eandp.com.au	+61 2 8070 6671
Sam Fletcher	Managing Director	Institutional Equities Sales	sam.fletcher@eandp.com.au	+61 2 8070 6615
Anna Lenzner	Executive Director	Institutional Equities Sales	anna.lenzner@eandp.com.au	+61 2 8070 6652
Steven Marchio	Executive Director	Institutional Equities Sales	steven.marchio@eandp.com.au	+61 3 9235 9752
Bill Forde	Executive Director	Institutional Equities Sales	bill.forde@eandp.com.au	+61 2 8070 6626
James O'Hare	Executive Director	Head of Fixed Income	james.ohare@eandp.com.au	+61 3 9631 9891
Tanya Irivine	Executive Director	Head of Corporate Access	tanya.irvine@eandp.com.au	+61 2 8070 6612
Stuart Gordon	Director	Head of International Sales Trading	stuart.gordon@eandp.com.au	+61 2 8070 6684
Alexander Blight	Director	Sales Trader	alexander.blight@eandp.com.au	+61 2 8070 6629
James Helliwell	Director	Fixed Income	james.helliwell@eandp.com.au	+61 2 8070 6614
Simon Bernadou	Associate Director	Institutional DTR	simon.bernadou@eandp.com.au	+61 2 8070 6672
Mark Steer	Associate Director	Institutional Operator	mark.steer@eandp.com.au	+61 3 9235 9708
Taylor Duong	Associate Director	Institutional Middle Office	taylor.duong@eandp.com.au	+61 2 8662 9722
Stefan Cvetkovic	Associate Director	Bond Desk Associate	stefan.cvetkovic@eandp.com.au	+61 3 9235 9731
Czarlette Cheika	Associate	Corporate Access Specialist	czarlette.cheika@eandp.com.au	+61 2 8070 6622
Paris Gutierrez	Associate	Corporate Access Specialist	paris.gutierrez@eandp.com.au	+61 2 8070 6611
Stephanie Siomos	Analyst	Fixed Income	stephanie.siomos@eandp.com.au	+61 3 9631 9887
Anji Pandey		Team Assistant - Equities	anji.pandey@eandp.com.au	+61 2 8662 9751
INSTITUTIONAL SALES (HONG KONG)				
Olivia Cartwright	Director	Australian Equities Sales	olivia.cartwright@eandp.com.au	+852 3899 6673

Office locations

AUSTRALIA – EVANS AND PARTNERS PTY LTD

Sydney

Level 32, 1 O'Connell Street

Sydney NSW 2000

T +61 2 8070 6600

F +61 2 8569 0386

W eandp.com.au

Melbourne

Mayfair Building, 171 Collins Street

Melbourne VIC 3000

T +61 3 9631 9888

F +61 3 8610 1608

HONG KONG – E&P ASIA (HK) LIMITED

Hong Kong

Suite 2076, Level 20, One International Finance Centre, No. 1 Harbour View Street

Central, Hong Kong

T +852 3899 6673

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